

# Chapter 12: Ratios, rates and time

11 lessons · one lesson for each exercise · each lesson is one hour

**NSW syllabus outcome**

Stage 4 Number and algebra: MA4-RAT-C-01 solves problems involving ratios and rates, and analyses distance–time graphs; time calculations from Stage 3 extended.

NAME \_\_\_\_\_

CLASS \_\_\_\_\_

CHAPTER

# 12

**12.01** Ratios

**12.02** Equivalent and simplified ratios

**12.03** Ratio problems

**12.04** Rates

**12.05** Best buys

**12.06** Rate problems

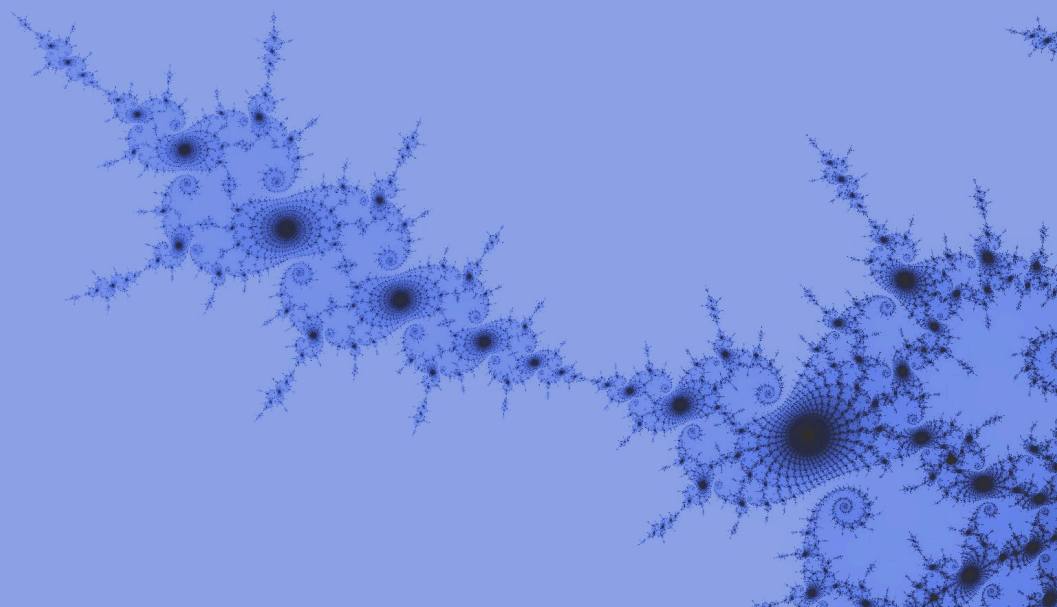
**12.07** Travel graphs

**12.08** Time

**12.09** 12-hour and 24-hour time

**12.10** Time differences

**12.11** Timetables



## How to use this booklet

**1** Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

**WE DO** **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

**2** Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

**YOU DO** **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

**DRILL** **Skill drill pages.** After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

**3** Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

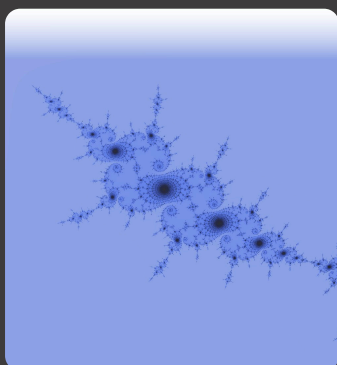
**4** Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

## Contents and progress tracker

| Lesson       | Topic                            | Page      | Date | Done                     |
|--------------|----------------------------------|-----------|------|--------------------------|
| <b>12.01</b> | Ratios                           | <b>3</b>  |      | <input type="checkbox"/> |
| <b>12.02</b> | Equivalent and simplified ratios | <b>7</b>  |      | <input type="checkbox"/> |
| <b>12.03</b> | Ratio problems                   | <b>11</b> |      | <input type="checkbox"/> |
| <b>12.04</b> | Rates                            | <b>15</b> |      | <input type="checkbox"/> |
| <b>12.05</b> | Best buys                        | <b>19</b> |      | <input type="checkbox"/> |
| <b>12.06</b> | Rate problems                    | <b>23</b> |      | <input type="checkbox"/> |
| <b>12.07</b> | Travel graphs                    | <b>27</b> |      | <input type="checkbox"/> |
| <b>12.08</b> | Time                             | <b>33</b> |      | <input type="checkbox"/> |
| <b>12.09</b> | 12-hour and 24-hour time         | <b>37</b> |      | <input type="checkbox"/> |
| <b>12.10</b> | Time differences                 | <b>41</b> |      | <input type="checkbox"/> |
| <b>12.11</b> | Timetables                       | <b>45</b> |      | <input type="checkbox"/> |

## In this chapter you will

- write, simplify and use ratios
- solve ratio problems
- write rates and find the best buy
- solve rate problems and read travel graphs
- round, convert and calculate with time
- use 12-hour and 24-hour time, time differences and timetables



### About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule,  $z \rightarrow z^2 + c$ , thousands of times for every point. Every booklet shows a different part of the set: this one is a jellyfish of spirals near the neck of the left bulb. Maths can make beautiful things.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON  
12.01

# Ratios

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- write a ratio to compare quantities

## Success criteria

I can:

- ☐ write a ratio in the order the words give
- ☐ use a colon: 3 : 5
- ☐ find a part-to-part and a part-to-whole ratio

**Key terms** Write the definition as your teacher explains it.

**Ratio**

**Part**

**Whole**

Notes

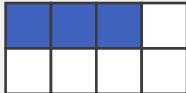
## Level 1 · Getting started

Easy · 28 questions

**WE DO** With your teacher: copy what they write on the board.

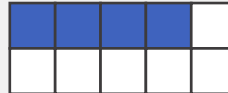
**Example 1** Write the ratio of shaded to unshaded parts.

**a** Ratio of shaded to unshaded?



Your teacher works this out. Copy every step.

**b** Ratio of shaded to unshaded?



Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

**REMEMBER** A ratio compares quantities of the same kind.

**ROUND 1** Questions 1–12: write the ratio (do not simplify).

① 2 red, 2 yellow: yellow to red

② 7 blue, 1 green: green to blue

③ 4 red, 15 blue: red to blue

④ 10 green, 5 blue: blue to green

⑤ 5 blue, 5 yellow: yellow to blue

⑥ 10 green, 8 blue: blue to green

⑦ 15 red, 3 yellow: yellow to red

⑧ 1 red, 15 blue: blue to red

⑨ 7 yellow, 15 green: yellow to green

⑩ 8 yellow, 3 red: red to yellow

⑪ 5 blue, 13 red: blue to all

⑫ 5 yellow, 5 red: yellow to red

**YOU DO** On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

**ROUND 2** Questions 13–28: write the ratio in simplest form.**13** 21 girls to 18 boys

---

**14** 16 girls to 4 boys

---

**15** 8 girls to 12 boys

---

**16** 8 girls to 18 boys

---

**17** 10 girls to 35 boys

---

**18** 25 girls to 5 boys

---

**19** 6 girls to 6 boys

---

**20** 16 girls to 18 boys

---

**21** 40 girls to 5 boys

---

**22** 12 girls to 30 boys

---

**23** 10 girls to 8 boys

---

**24** 3 girls to 27 boys

---

**25** 12 girls to 36 boys

---

**26** 36 girls to 42 boys

---

**27** 8 girls to 8 boys

---

**28** 18 girls to 15 boys

---

**Level 2 · Building up**

Medium · 18 questions

**WE DO** With your teacher: copy what they write on the board.**Example 2** Write each ratio in the order the words give.**a** 17 women and 38 men: find the ratio of men to women.

Your teacher works this out. Copy every step.

**b** Find the ratio of women to the total.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

18 QUESTIONS

**REMEMBER** Write the parts in the order given.**ROUND 1** Questions 1–12: change to the same unit, then write the ratio in simplest form.**1** 40 c to 1 \$

---

**2** 20 mm to 3 cm

---

**3** 10 mm to 3 cm

---

**4** 10 min to 1 h

---

**5** 70 min to 1 h

---

**6** 20 g to 4 kg

---

**7** 40 cm to 3 m

---

**8** 40 min to 2 h

---

**9** 90 mm to 4 cm

---

**10** 10 min to 5 h

---

**11** 20 min to 1 h

---

**12** 100 c to 4 \$

---

**YOU DO** On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

**ROUND 2** Questions 13–18: write the ratio of all three in simplest form.

13 25 red, 5 blue, 30 green

\_\_\_\_\_

14 4 red, 4 blue, 12 green

\_\_\_\_\_

15 20 red, 10 blue, 25 green

\_\_\_\_\_

16 8 red, 8 blue, 12 green

\_\_\_\_\_

17 10 red, 10 blue, 6 green

\_\_\_\_\_

18 5 red, 5 blue, 10 green

\_\_\_\_\_

**Level 3 · Challenge**

Challenging · 3 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Write the ratio, then explain.

**a** A recipe uses 2 cups of flour and 1 cup of sugar.  
Write the ratio of flour to sugar.

\_\_\_\_\_

Your teacher works this out. Copy every step.

**b** Is 3 : 5 the same as 5 : 3? Explain.

\_\_\_\_\_

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Think it through.

3 QUESTIONS

**REMEMBER** Use a colon: 3 : 5 (read “3 to 5”).

1 A drink is 1 part cordial to 4 parts water. What fraction is cordial?

\_\_\_\_\_

2 In a ratio of cats to dogs of 3 : 2, there are 12 cats. How many dogs?

\_\_\_\_\_

3 Draw a 10-square diagram with a shaded to unshaded ratio of 3 : 2.

\_\_\_\_\_

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

6 cats and 5 dogs. Write the ratio of cats to dogs.

\_\_\_\_\_

**Level 2**

Write the ratio of dogs to all the animals.

\_\_\_\_\_

**Level 3**

A juice is 1 part concentrate to 6 parts water. Write the ratio of water to the whole drink.

\_\_\_\_\_

 Extension · Stretch yourself Worked example

The ratio of boys to girls is 4 : 5. What fraction of the class are boys?

1. Total parts =  $4 + 5 = 9$
2. Boys are 4 of the 9 parts

Answer:  $\frac{4}{9}$

**YOU DO**

On your own: read the worked example, then try these.

- E1** The ratio of red to blue to green beads is 2 : 3 : 5. What fraction are blue?

- E2** In a 3 : 7 ratio of juice to water, what percentage is juice?

**Lesson summary**

Missed the lesson? Start here.

- Remember**
1. A ratio compares quantities of the same kind.
  2. Write the parts in the order given.
  3. Use a colon: 3 : 5 (read “3 to 5”).
  4. Part to whole: add the parts for the whole.

 Level 1

3 shaded, 5 unshaded

1. shaded : unshaded

Answer: 3 : 5

 Level 2

Men to women

1. 38 : 17

Answer: 38 : 17

 Level 3

Women to total

1. 17 : 55

Answer: 17 : 55

 Extension

Boys : girls = 4 : 5

1. 4 of 9 parts

Answer:  $\frac{4}{9}$  are boys

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
12.02

# Equivalent and simplified ratios

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find equivalent ratios and simplify ratios, including with units

## Success criteria

I can:

- ☐ multiply or divide every part by the same number
- ☐ divide by the HCF to simplify
- ☐ change to the same units before simplifying

**Key terms** Write the definition as your teacher explains it.

**Equivalent ratio**

**Simplest form**

Notes

## Level 1 · Getting started

Easy · 32 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Complete each pair of equivalent ratios.

**a** Complete:  $2 : 3 = 8 : \underline{\hspace{1cm}}$

**b** Complete:  $2 : 3 : 4 = 6 : \underline{\hspace{1cm}} : \underline{\hspace{1cm}}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

**REMEMBER** Equivalent:  $\times$  or  $\div$  every part by the same number.

**ROUND 1** Questions 1–16: simplify the ratio.

**1**  $42 : 48$

**2**  $8 : 36$

**3**  $22 : 6$

**4**  $12 : 28$

**5**  $45 : 36$

**6**  $66 : 42$

**7**  $14 : 10$

**8**  $24 : 22$

**9**  $3 : 15$

**10**  $8 : 88$

**11**  $10 : 18$

**12**  $55 : 5$

**13**  $36 : 42$

**14**  $40 : 96$

**15**  $8 : 4$

**16**  $14 : 63$

**YOU DO**

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

**ROUND 2**

Questions 17–32: find the missing number.

17  $2 : 2 = \square : 8$

\_\_\_\_\_

18  $6 : 3 = 36 : \square$

\_\_\_\_\_

19  $8 : 7 = \square : 56$

\_\_\_\_\_

20  $6 : 6 = \square : 48$

\_\_\_\_\_

21  $7 : 2 = 49 : \square$

\_\_\_\_\_

22  $3 : 9 = 21 : \square$

\_\_\_\_\_

23  $1 : 7 = \square : 21$

\_\_\_\_\_

24  $9 : 8 = 36 : \square$

\_\_\_\_\_

25  $7 : 9 = \square : 36$

\_\_\_\_\_

26  $8 : 6 = \square : 12$

\_\_\_\_\_

27  $2 : 1 = 4 : \square$

\_\_\_\_\_

28  $8 : 6 = 32 : \square$

\_\_\_\_\_

29  $6 : 8 = 18 : \square$

\_\_\_\_\_

30  $1 : 9 = \square : 63$

\_\_\_\_\_

31  $1 : 8 = \square : 64$

\_\_\_\_\_

32  $3 : 2 = \square : 12$

\_\_\_\_\_



**Level 2 · Building up**

Medium · 22 questions

**WE DO**

With your teacher: copy what they write on the board.

**Example 2**

Simplify each ratio.

**a** Simplify  $12 : 21$ .

Your teacher works this out. Copy every step.

**b** Simplify  $18 : 6 : 24$ .

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

**REMEMBER** Simplify:  $\div$  by the HCF of all parts.

**ROUND 1**

Questions 1–16: simplify. Change decimals or fractions to whole numbers first.

1  $0.3 : 0.7$

\_\_\_\_\_

2  $0.9 : 0.4$

\_\_\_\_\_

3  $0.5 : 0.1$

\_\_\_\_\_

4  $0.5 : 0.6$

\_\_\_\_\_

5  $0.2 : 0.5$

\_\_\_\_\_

6  $0.8 : 0.4$

\_\_\_\_\_

7  $0.6 : 0.4$

\_\_\_\_\_

8  $0.8 : 0.3$

\_\_\_\_\_

9  $0.9 : 0.8$

\_\_\_\_\_

10  $0.5 : 0.8$

\_\_\_\_\_

11  $0.9 : 0.2$

\_\_\_\_\_

12  $0.8 : 0.2$

\_\_\_\_\_

13  $0.6 : 0.3$

\_\_\_\_\_

14  $0.4 : 0.2$

\_\_\_\_\_

15  $0.7 : 0.5$

\_\_\_\_\_

16  $0.6 : 0.1$

\_\_\_\_\_

**YOU DO**

On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

## ROUND 2

**Questions 17–22: simplify the three-part ratio.**

17 2 : 2 : 8

18  $3:6:3$

19 21:14:28

20 3 : 21 : 3

21  $48 : 8 : 8$

22 2 : 10 : 12



### Level 3 • Challenge

Challenging · 3 questions · then try the extension

**WE DO**

**With your teacher: copy what they write on the board.**

### Example 3

**Change to the same units, then simplify.**

**a** Simplify 2 cm : 10 mm.

**b** Simplify 20 minutes : 1 hour.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Challenging** Simplify.

### 3 QUESTIONS

**REMEMBER** Units: change to the same unit first, then drop the units.

① 20 minutes : 1 hour

② 12 days : 3 weeks

③ 4 days : 10 hours

**YOU DO**

## Show off your skill

Choose **ONE** question. Show your working.

| Name | Class |
|------|-------|
|------|-------|



Complete:  $3 : 7 = 9 : \underline{\hspace{1cm}}$



Simplify 16 : 40.



Simplify  $30 \text{ cm} : 2 \text{ m}$ .

**Extension · Stretch yourself**

**Worked example**

Simplify  $\frac{1}{2} : \frac{1}{3}$ .

1. Multiply both by 6 (the LCD)
2.  $3 : 2$

**Answer:**  $3 : 2$

**YOU DO**

On your own: read the worked example, then try these.

**E1** Simplify  $\frac{3}{4} : \frac{5}{8}$ .

**E2** Simplify  $0.25 \text{ kg} : 750 \text{ g} : 1.5 \text{ kg}$ .

**Lesson summary**

Missed the lesson? Start here.

**Remember**

1. Equivalent:  $\times$  or  $\div$  every part by the same number.
2. Simplify:  $\div$  by the HCF of all parts.
3. Units: change to the same unit first, then drop the units.
4. Decimals or fractions: multiply to make whole numbers.

**Level 1**

$2 : 3 = 8 : ?$

1.  $\times 4$

**Answer:**  $8 : 12$

**Level 2**

$12 : 21$

1.  $\div 3$

**Answer:**  $4 : 7$

**Level 3**

$2 \text{ cm} : 10 \text{ mm}$

1.  $20 \text{ mm} : 10 \text{ mm}$

**Answer:**  $2 : 1$

**Extension**

$\frac{1}{2} : \frac{1}{3}$

1.  $\times 6$

**Answer:**  $3 : 2$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
12.03

# Ratio problems

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- solve problems using ratios

## Success criteria

I can:

- ☐ use the unitary method: find one part first
- ☐ find an unknown quantity from a ratio
- ☐ divide a quantity in a given ratio

**Key terms** Write the definition as your teacher explains it.

### Unitary method

### Share

### Notes

## Level 1 · Getting started

Easy · 20 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Use the ratio to find the unknown amount.

- a** Girls : boys = 4 : 3. There are 75 boys. How many girls?

- b** Tyrone : Erin invested in the ratio 5 : 7. Erin invested \$63 000. How much did Tyrone invest?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **20 QUESTIONS**

**REMEMBER** Find the value of one part.

**ROUND 1** Questions 1–12: share the amount in the ratio given.

**1** \$35 in 4 : 3

**2** \$55 in 4 : 1

**3** \$90 in 3 : 2

**4** \$12 in 1 : 3

**5** \$90 in 1 : 5

**6** \$20 in 4 : 1

**7** \$44 in 3 : 1

**8** \$30 in 1 : 4

**9** \$56 in 3 : 4

**10** \$24 in 2 : 1

**11** \$120 in 4 : 2

**12** \$60 in 1 : 3



**YOU DO** On your own: **Medium** keep going: Round 2.

4 QUESTIONS MORE

**ROUND 2** Questions 13–16: the difference between the two shares is given. Find each share.

13 ratio 4 : 7; the larger share is \$33 more

\_\_\_\_\_

14 ratio 2 : 4; the larger share is \$6 more

\_\_\_\_\_

15 ratio 4 : 5; the larger share is \$12 more

\_\_\_\_\_

16 ratio 4 : 5; the larger share is \$5 more

\_\_\_\_\_

**Level 3 · Challenge**

Challenging · 3 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Solve the ratio problem.

**a** A map has scale 1 : 50 000. 3 cm on the map is how far in real life?

\_\_\_\_\_

Your teacher works this out. Copy every step.

**b** Cement, sand and gravel are mixed 1 : 2 : 3. How much sand is in 36 kg of mix?

\_\_\_\_\_

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Harder problems.

3 QUESTIONS

**REMEMBER** To share: total parts, then amount ÷ total parts.

1 The ratio of adults to children at a show is 3 : 8. There were 440 people. How many were children?

\_\_\_\_\_

2 Sam and Jo share money 3 : 5. Jo gets \$24 more than Sam. How much does each get?

\_\_\_\_\_

3 A recipe for 4 people uses 300 g of pasta. How much for 10 people?

\_\_\_\_\_

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Complete:  $2 : 7 = 10 : \underline{\hspace{1cm}}$

\_\_\_\_\_

**Level 2**

Share \$80 in the ratio 3 : 5.

\_\_\_\_\_

**Level 3**

Juice : water = 1 : 4. How much water for 250 mL of juice?

\_\_\_\_\_

 Extension · Stretch yourself Worked example

The ratio of red to blue marbles is 3 : 5. After 6 red marbles are added, the ratio is 1 : 1. How many blue marbles are there?

1. Red  $3k$ , blue  $5k$
2.  $3k + 6 = 5k$ , so  $2k = 6$ ,  $k = 3$

Answer: 15 blue

## YOU DO

On your own: read the worked example, then try these.

- E1 The ratio of boys to girls is 2 : 3. After 4 boys join, it is 1 : 1. How many girls?

- E2 Share \$84 among A, B and C so that A gets twice B and C gets three times B.

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Find the value of one part.
  2. Multiply to find the other quantities.
  3. To share: total parts, then amount  $\div$  total parts.
  4. Check the parts add back to the total.

 Level 1

Girls : boys = 4 : 3, boys 75

1. 1 part = 25

Answer: 100 girls

 Level 2

Tyrone : Erin = 5 : 7

1. 1 part =  $63\,000 \div 7 = 9000$

Answer: \$45 000

 Level 3

\$450 in 4 : 5

1. 9 parts, 1 part = \$50

Answer: \$200 and \$250

 Extension

Marbles 3 : 5  $\rightarrow$  1 : 1

1.  $2k = 6$

Answer: 15 blue

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
12.04

# Rates

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- write and simplify rates

## Success criteria

I can:

- ☐ know a rate compares quantities of different kinds
- ☐ write a rate with both units, e.g. km/h
- ☐ simplify a rate to a rate per 1 unit

**Key terms** Write the definition as your teacher explains it.

**Rate**

**Unit rate**

Notes

## Level 1 · Getting started

Easy · 24 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Write each as a rate in simplest form.

- a** Write as a rate: 150 000 voters for 50 politicians.

Your teacher works this out. Copy every step.

- b** Write as a rate: 1000 sheep over 5 hectares.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS**

**REMEMBER** A **RATE** compares different kinds of quantities.

**ROUND 1** Questions 1–12: write as a rate for one unit.

- |                               |                              |                                |
|-------------------------------|------------------------------|--------------------------------|
| 1 \$165 in 3 kg<br>_____      | 2 72 words in 4 min<br>_____ | 3 15 words in 5 min<br>_____   |
| 4 156 beats in 6 min<br>_____ | 5 35 L in 5 min<br>_____     | 6 306 words in 6 min<br>_____  |
| 7 \$108 in 9 kg<br>_____      | 8 18 km in 3 h<br>_____      | 9 133 m in 7 s<br>_____        |
| 10 432 km in 8 h<br>_____     | 11 114 L in 3 min<br>_____   | 12 240 beats in 4 min<br>_____ |

## YOU DO

On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

## ROUND 2

Questions 13–24: what unit would you use for each rate?

13 typing speed

---

14 running speed

---

15 wage

---

16 growth of a plant

---

17 petrol price

---

18 phone data

---

19 water from a tap

---

20 price of apples

---

21 rainfall

---

22 fuel use

---

23 heart rate

---

24 speed of a car

---



## Level 2 · Building up

Medium · 16 questions

## WE DO

With your teacher: copy what they write on the board.

## Example 2

Write each as a rate in simplest form.

a Write as a rate: 10 tickets for \$8.

Your teacher works this out. Copy every step.

b Write as a rate: 810 stitches to make 3 balls.

Your teacher works this out. Copy every step.

## YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

16 QUESTIONS

**REMEMBER** Write both units: km/h, \$/kg, L/min.

## ROUND 1

Questions 1–12: convert the rate.

1 6 km/h to km/min

---

2 120 km/h to km/min

---

3 120 km/min to km/h

---

4 30 km/min to km/h

---

5 \$36/kg to c/g

---

6 18 km/h to km/min

---

7 \$60/kg to c/g

---

8 90 km/min to km/h

---

9 36 km/h to km/min

---

10 60 km/min to km/h

---

11 \$6/kg to c/g

---

12 90 km/h to km/min

---

**YOU DO**

On your own: **Medium** keep going: Round 2.

4 QUESTIONS MORE

## ROUND 2

**Questions 13–16: which rate is faster (or cheaper)? Change both to the same unit.**

- 13** A: 47 km/h, B: 394 km in 4 h

- 14 A: 44 km/h, B: 163 km in 5 h

- 15** A: 36 km/h, B: 253 km in 4 h

- 16 A: 53 km/h, B: 145 km in 4 h



### Level 3 · Challenge

Challenging · 3 questions · then try the extension

**WE DO**

**With your teacher: copy what they write on the board.**

### Example 3

**Find the speed, then convert the rate.**

- a** A car travels 240 km in 3 hours. Find the speed in km/h.

- b** Convert 90 km/h to metres per minute.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Challenging** Converting rates.

### 3 QUESTIONS

## REMEMBER

Unit rate: divide to get “per 1”.

- 1 Convert 90 km/h to metres per minute.

- 2 Convert 5 m/s to km/h.

- 3 Convert \$18 per hour to cents per minute.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO**

## Show off your skill

Choose **ONE** question. Show your working.

| Name | Class   |
|------|---------|
| John | Math    |
| Jane | Math    |
| John | Science |
| Jane | Science |
| John | History |
| Jane | History |
| John | Art     |
| Jane | Art     |
| John | Music   |
| Jane | Music   |



Write \$36 for 4 kg as a rate per kg.



A car travels 150 km in 2 h. Find its speed.



Convert 60 km/h to metres per minute.

 Extension · Stretch yourself Worked example

Convert 72 km/h to m/s.

1.  $72 \text{ km} = 72\,000 \text{ m}$ ;  $1 \text{ h} = 3600 \text{ s}$
2.  $72\,000 \div 3600$

Answer: 20 m/s

## YOU DO

On your own: read the worked example, then try these.

- E1 Convert 15 m/s to km/h.

- E2 A snail moves at 3 mm/s. How many metres per hour is that?

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. A **rate** compares different kinds of quantities.
  2. Write both units: km/h, \$/kg, L/min.
  3. Unit rate: divide to get “per 1”.
  4. Convert units on the top and bottom separately.

 Level 1

150 000 voters, 50 politicians

1.  $\div 50$

Answer: 3000 voters/politician

 Level 2

10 tickets for \$8

1.  $8 \div 10$

Answer: \$0.80/ticket

 Level 3

240 km in 3 h

1.  $240 \div 3$

Answer: 80 km/h

 Extension

72 km/h to m/s

1.  $72\,000 \div 3600$

Answer: 20 m/s

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
12.05

## Best buys

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- compare prices to find the best buy

## Success criteria

I can:

- ☐ find the unit price (price per 1 unit or per 100 g)
- ☐ change to the same units before comparing
- ☐ decide the best buy and give a reason

**Key terms** Write the definition as your teacher explains it.**Unit price****Best buy**

Notes

## Level 1 · Getting started

Easy · 20 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Find the price per 100 g, then choose the better buy.

- a** Dog biscuits: 400 g of Woof for \$5.20. Find the price per 100 g.

Your teacher works this out. Copy every step.

- b** 1 kg of Chow for \$11.10. Find the price per 100 g. Which is the better buy?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **20 QUESTIONS****REMEMBER** Find the price for the same amount (per 1 or per 100 g).**ROUND 1** Questions 1–12: find the unit price (the cost of one).**1** 7 for \$26.18**2** 4 for \$35.00**3** 4 for \$8.92**4** 10 for \$70.60**5** 3 for \$21.12**6** 5 for \$5.55**7** 3 for \$19.02**8** 3 for \$11.43**9** 6 for \$51.78**10** 11 for \$68.31**11** 8 for \$68.48**12** 6 for \$22.68

**YOU DO** On your own: **Easy** keep going: Round 2.

8 QUESTIONS MORE

**ROUND 2** Questions 13–20: which is the better buy? Find the price of 1 of each.**13** 4 for \$15.56 or 3 for \$11.94**14** 3 for \$11.70 or 5 for \$5.10**15** 6 for \$23.40 or 2 for \$4.68**16** 2 for \$5.70 or 3 for \$9.48**17** 3 for \$10.02 or 5 for \$12.75**18** 2 for \$3.44 or 6 for \$6.84**19** 4 for \$15.08 or 3 for \$9.06**20** 4 for \$4.08 or 6 for \$8.52**Level 2 · Building up**

Medium · 14 questions

**WE DO** With your teacher: copy what they write on the board.**Example 2** Which is the better buy?**a** Milk: 1 L of Dairy Farm for \$2.85 or 1250 mL of Milky for \$3.90. Which is better value?

Your teacher works this out. Copy every step.

**b** Juice: 2 L for \$4.60 or 3 L for \$6.60. Which is better value?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

14 QUESTIONS

**REMEMBER** Change units first (1 kg = 1000 g, 1 L = 1000 mL).**ROUND 1** Questions 1–8: which is the better buy? Find the price per 100 g (or per 100 mL).**1** 375 g for \$5.44 or 750 g for \$5.72**2** 150 g for \$5.96 or 600 g for \$9.28**3** 375 g for \$2.90 or 750 g for \$6.07**4** 250 g for \$1.67 or 500 g for \$9.41**5** 150 g for \$1.90 or 600 g for \$7.16**6** 250 g for \$3.00 or 500 g for \$7.56**7** 150 g for \$5.67 or 600 g for \$11.40**8** 300 g for \$3.66 or 1000 g for \$7.53

**YOU DO** On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

**ROUND 2** Questions 9–14: find the cost.

9 0.5 kg at \$5.36/kg

10 0.5 kg at \$13.26/kg

11 1.5 kg at \$10.24/kg

12 2.5 kg at \$2.72/kg

13 6 kg at \$7.73/kg

14 5 kg at \$8.97/kg

**Level 3 · Challenge**

Challenging · 3 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Which is the better buy? Explain.

**a** Why might someone not choose the best buy?

**b** Pens: 3 for \$4.20 or 5 for \$6.50. Which is cheaper per pen?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Problems.

3 QUESTIONS

**REMEMBER** The lower unit price is the better buy.

1 Rice: 1 kg for \$2.40, 2 kg for \$4.50, 5 kg for \$10.50. Which is the best buy?

2 A 600 g jar costs \$4.20. A 1 kg jar costs \$6.50. How much do you save per 100 g with the bigger jar?

3 Is buying in bulk always the best choice? Explain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill** Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Find the price per kg: 3 kg for \$12.60.

**Level 2**

Which is better: 1 L for \$3 or 2 L for \$5.40?

**Level 3**

Which is better: 250 g for \$2.50 or 1 kg for \$9?

 Extension · Stretch yourself Worked example

A shop has “3 for \$10” or \$3.60 each. How much do you save buying 6?

1. 6 at \$3.60 = \$21.60
2. 2 lots of 3 for \$10 = \$20

Answer: \$1.60

**YOU DO**

On your own: read the worked example, then try these.

- E1** Shampoo: 400 mL for \$6 or 1 L for \$13.50. Which is better? By how much per 100 mL?

- E2** A “buy 2 get 1 free” deal on \$4.50 items: find the price per item.

**Lesson summary**

Missed the lesson? Start here.

- Remember**
1. Find the price for the same amount (per 1 or per 100 g).
  2. Change units first (1 kg = 1000 g, 1 L = 1000 mL).
  3. The lower unit price is the better buy.
  4. Consider waste, storage and need.

 Level 1

Woof

1.  $5.20 \div 4$

Answer: \$1.30/100 g

 Level 2

Chow

1.  $11.10 \div 10$

Answer: \$1.11/100 g: better

 Level 3

Milk

1. \$2.85/L vs \$3.12/L

Answer: Dairy Farm

 Extension

3 for \$10 vs \$3.60 each (6)

1. \$20 vs \$21.60

Answer: save \$1.60

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
12.06

# Rate problems

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- solve problems using rates

## Success criteria

I can:

- ☐ multiply a rate by a quantity to find a total
- ☐ divide by a rate to find how many
- ☐ round sensibly

**Key terms** Write the definition as your teacher explains it.

**Rate**

**Unitary method**

Notes

## Level 1 · Getting started

Easy · 28 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Use the rate to find the total.

- a** Marie earns \$22.45 per hour. How much for 38 hours?

- b** Petrol costs \$1.79/L. How many litres can you buy for \$65 (1 d.p.)?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

**REMEMBER** Total = rate × amount.

**ROUND 1** Questions 1–12: use the rate to find the total.

**1** 6 kg at \$25/kg

**2** 3 kg at \$12/kg

**3** 11 kg at \$19/kg

**4** 4 h at 70 km/h

**5** 5 kg at \$15/kg

**6** 7 kg at \$16/kg

**7** 6 kg at \$23/kg

**8** 9 h at 105 km/h

**9** 4 h at \$20/h

**10** 2 h at \$11/h

**11** 3 h at \$4/h

**12** 9 h at \$16/h

**YOU DO**

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

## ROUND 2

**Questions 13–28: find the time taken:  $\text{time} = \text{distance} \div \text{speed}$ .**

- |       |                    |       |                    |       |                    |       |                   |
|-------|--------------------|-------|--------------------|-------|--------------------|-------|-------------------|
| 13    | 400 km at 80 km/h  | 14    | 200 km at 50 km/h  | 15    | 100 km at 50 km/h  | 16    | 250 km at 50 km/h |
| <hr/> |                    | <hr/> |                    | <hr/> |                    | <hr/> |                   |
| 17    | 160 km at 80 km/h  | 18    | 120 km at 40 km/h  | 19    | 150 km at 50 km/h  | 20    | 80 km at 80 km/h  |
| <hr/> |                    | <hr/> |                    | <hr/> |                    | <hr/> |                   |
| 21    | 60 km at 60 km/h   | 22    | 300 km at 100 km/h | 23    | 300 km at 50 km/h  | 24    | 320 km at 80 km/h |
| <hr/> |                    | <hr/> |                    | <hr/> |                    | <hr/> |                   |
| 25    | 200 km at 100 km/h | 26    | 180 km at 60 km/h  | 27    | 500 km at 100 km/h | 28    | 240 km at 40 km/h |
| <hr/> |                    | <hr/> |                    | <hr/> |                    | <hr/> |                   |

 **Level 2 · Building up**

Medium · 15 questions

**WE DO**

**With your teacher: copy what they write on the board.**

### Example 2

**Use the rate to find how many.**

- a** Fertiliser is used at 20 kg/ha. How many hectares does 150 kg cover?
- b** A car travels at 80 km/h for 2.5 h. How far does it go?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line (show working in the boxes).

## 15 QUESTIONS

**REMEMBER** How many = total  $\div$  rate.

## ROUND 1

**Questions 1–12: find the distance: distance = speed  $\times$  time.**

- |                           |                           |                           |
|---------------------------|---------------------------|---------------------------|
| 1 60 km/h for 30 min      | 2 90 km/h for 15 min      | 3 60 km/h for 1 h 30 min  |
| 4 100 km/h for 30 min     | 5 90 km/h for 45 min      | 6 80 km/h for 30 min      |
| 7 60 km/h for 45 min      | 8 120 km/h for 2 h 30 min | 9 90 km/h for 1 h 30 min  |
| 10 80 km/h for 1 h 30 min | 11 40 km/h for 1 h 30 min | 12 90 km/h for 2 h 30 min |

**YOU DO** On your own: **Medium** keep going: Round 2.

3 QUESTIONS MORE

**ROUND 2** Questions 13–15: two steps. Show your working.

- 13 Mia earns \$29/h. She works 6 h a day for 5 days. How much does she earn?

- 14 Mia earns \$15/h. She works 8 h a day for 5 days. How much does she earn?

- 15 Mia earns \$26/h. She works 3 h a day for 5 days. How much does she earn?

**Level 3 · Challenge**

Challenging · 3 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Use the rate to find the time or distance.

- a How long does it take to travel 210 km at 70 km/h?

Your teacher works this out. Copy every step.

- b A tap drips 12 mL per minute. How many litres in a day?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Harder problems.

3 QUESTIONS

**REMEMBER** Time = distance ÷ speed.

- 1 Two taps fill a tank: one at 20 L/min, the other at 30 L/min. How long to fill 1500 L together?

- 2 A car uses 8 L/100 km. Petrol is \$1.85/L. Find the fuel cost of a 450 km trip.

- 3 Ann types 45 words/min and Ben 60 words/min. How much longer does Ann take to type 1800 words?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** **Show off your skill** Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Find the pay for 6 h at \$24.50/h.

**Level 2**

How long to travel 180 km at 60 km/h?

**Level 3**

How many litres of petrol for \$50 at \$1.60/L?

 Extension · Stretch yourself Worked example

A train leaves at 9 a.m. travelling at 90 km/h. A car leaves the same place at 10 a.m. at 120 km/h on the same road. When does the car catch up?

1. At 10 a.m. the train is 90 km ahead
2. The car gains 30 km every hour
3.  $90 \div 30 = 3$  hours after 10 a.m.

Answer: 1 p.m.

## YOU DO

On your own: read the worked example, then try these.

- E1 How far from the start does the car catch up?

- E2 Two people walk towards each other from 12 km apart at 4 km/h and 2 km/h. When do they meet?

## Lesson summary

Missed the lesson? Start here.

## Remember

1. Total = rate  $\times$  amount.
2. How many = total  $\div$  rate.
3. Time = distance  $\div$  speed.
4. Check units match (h and min, L and mL).

 Level 1

$$\$22.45/\text{h} \times 38 \text{ h}$$

1.  $22.45 \times 38$

Answer: \$853.10

 Level 2

$$\$65 \div \$1.79/\text{L}$$

1. 36.31...

Answer: 36.3 L

 Level 3

$$150 \text{ kg} \div 20 \text{ kg/ha}$$

1. 7.5

Answer: 7.5 ha

 Extension

Car catches train

1.  $90 \div 30 = 3 \text{ h}$

Answer: 1 p.m.

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
12.07

# Travel graphs

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- read and interpret travel (distance–time) graphs

## Success criteria

I can:

- ☐ read distance and time from a travel graph
- ☐ identify stops (flat sections) and the return journey
- ☐ find speed = distance ÷ time for a section

**Key terms** Write the definition as your teacher explains it.

**Travel graph**

**Speed**

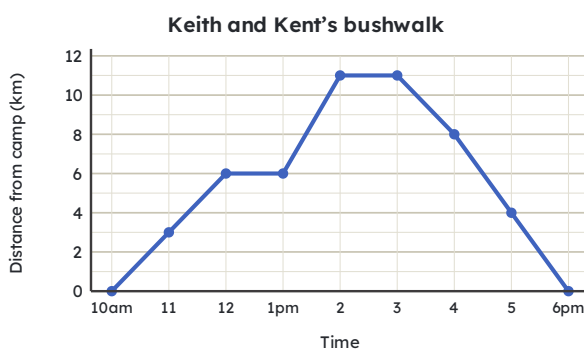
Notes

## Level 1 · Getting started

Easy · 12 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Use the travel graph.



**a** How far did Keith and Kent walk altogether?

Your teacher works this out. Copy every step.

**b** How many stops did they make?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **12 QUESTIONS**

**REMEMBER** Time is across, distance is up.

**ROUND 1** Questions 1–6: find the speed: speed = distance ÷ time.

1 75 km in 5 h

2 400 km in 4 h

3 20 km in 2 h

4 90 km in 2 h

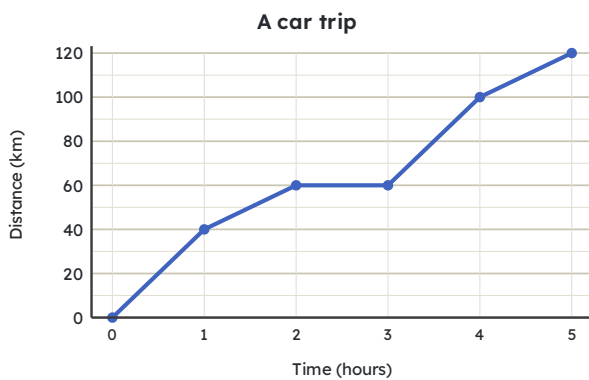
5 480 km in 6 h

6 200 km in 5 h

**YOU DO** On your own: **Easy** keep going: Round 2.

6 QUESTIONS MORE

**ROUND 2** Questions 7–12: use the travel graph.



7 How far was travelled in the first hour?

\_\_\_\_\_

8 What was the total distance?

\_\_\_\_\_

9 How long did the whole trip take?

\_\_\_\_\_

10 When did the traveller stop?

\_\_\_\_\_

11 In which hour was the speed greatest?

\_\_\_\_\_

12 What was the average speed for the whole trip?

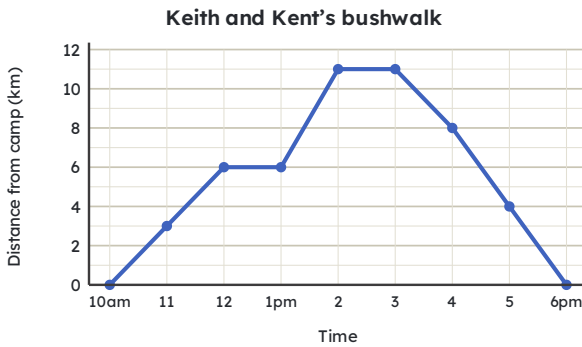
\_\_\_\_\_

**Level 2 · Building up**

Medium · 12 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Use the travel graph.



**a** At what time did they start back?

Your teacher works this out. Copy every step.

**b** Between what times did they walk fastest?

Your teacher works this out. Copy every step.

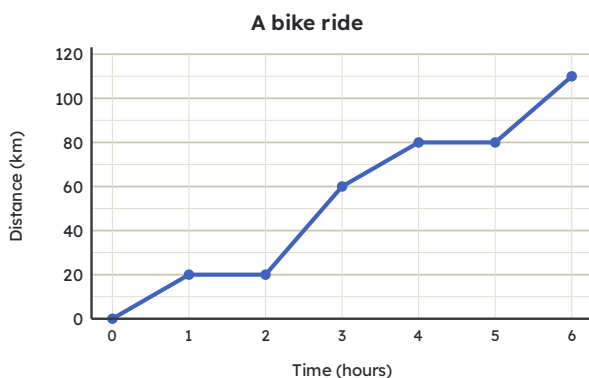
**YOU DO**

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

12 QUESTIONS

**REMEMBER** A flat line means stopped.

**ROUND 1** Questions 1–6: use the travel graph.



**1** How far was travelled in the first hour?

\_\_\_\_\_

**2** What was the total distance?

\_\_\_\_\_

**3** How long did the whole trip take?

\_\_\_\_\_

**4** When did the traveller stop?

\_\_\_\_\_

**5** In which hour was the speed greatest?

\_\_\_\_\_

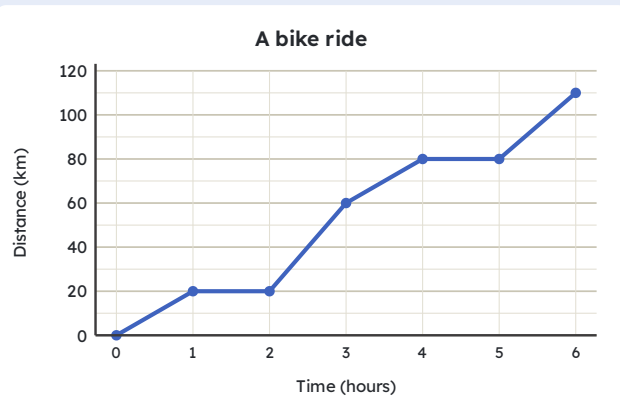
**6** What was the average speed for the whole trip?

\_\_\_\_\_

**YOU DO** On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

**ROUND 2** Questions 7–12: use the travel graph.



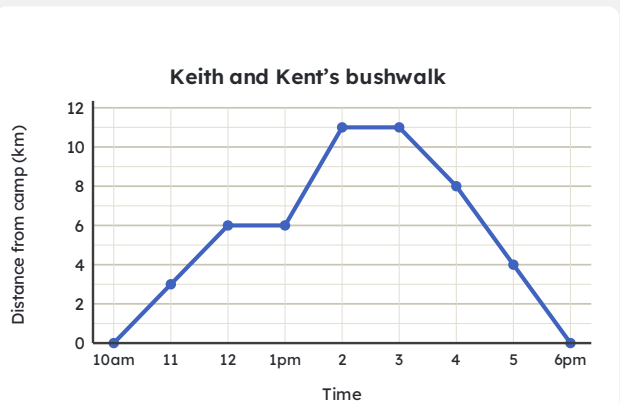
- 7 How far was travelled in the first hour?  
\_\_\_\_\_
- 8 What was the total distance?  
\_\_\_\_\_
- 9 How long did the whole trip take?  
\_\_\_\_\_
- 10 When did the traveller stop?  
\_\_\_\_\_
- 11 In which hour was the speed greatest?  
\_\_\_\_\_
- 12 What was the average speed for the whole trip?  
\_\_\_\_\_

**Level 3 · Challenge**

Challenging · 3 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Use the travel graph, then explain.



- a** Calculate their speed in the last 2 hours.

Your teacher works this out. Copy every step.

- b** What does a steeper line mean on a travel graph?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Think it through.

3 QUESTIONS

**REMEMBER** Steeper line means faster.

- 1 On a travel graph, what does a horizontal line show?
- 2 What does a line going down to the time axis show?
- 3 Write a story for Mia's car trip in Set B.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Two cyclists leave town A at 8 a.m. One rides at 20 km/h, the other at 25 km/h. How far apart are they at 10:30 a.m.?

1. Time = 2.5 h
2. Distances 50 km and 62.5 km

Answer: 12.5 km

YOU DO

On your own: read the worked example, then try these.

- E1 Draw both cyclists on one travel graph from 8 to 10:30 a.m.

- E2 When is the faster cyclist 10 km ahead?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

Level 1

What does a flat section on a travel graph mean?

Level 2

A graph goes from 0 km at 1 p.m. to 30 km at 3 p.m. Find the speed.

Level 3

In Set A, how far from camp were they at 11 a.m.?

## Lesson summary

Missed the lesson? Start here.

### Remember

1. Time is across, distance is up.
2. A flat line means stopped.
3. Steeper line means faster.
4.  $\text{Speed} = \text{change in distance} \div \text{change in time}$ .

#### Level 1

Total walked

1. 11 km out, 11 km back

Answer: 22 km

#### Level 2

Stops

1. Two flat sections

Answer: 2

#### Level 3

Last 2 hours

1.  $8 \text{ km} \div 2 \text{ h}$

Answer: 4 km/h

#### Extension

Cyclists at 10:30

1.  $62.5 - 50$

Answer: 12.5 km

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

## 12.08 Time

## LESSON 12.08

## Time

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- round, convert, add and subtract time

## Success criteria

I can:

- ☐ round time to the nearest hour or minute
- ☐ convert between hours, minutes and seconds
- ☐ add and subtract hours and minutes, carrying 60

**Key terms** Write the definition as your teacher explains it.

## Minute

## Second

## Notes

## Level 1 · Getting started

Easy · 24 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Round, then convert.

**a** Round 9 h 50 min to the nearest hour.

**b** Round 3.2 h to the nearest hour. How many minutes is 0.2 h?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

**REMEMBER** 60 s = 1 min, 60 min = 1 h, 24 h = 1 day.

**ROUND 1** Questions 1–12: convert. 1 h = 60 min, 1 min = 60 s.

① 2 h to min

② 6 min to s

3 380 min to h and min

4 480 min to h

5 300 min to h

⑥ 4 min to s

7 9 min to s

8 8 h to min

9 250 min to h and min

**10** 3 min to s

**11** 450 min to h and min

12 5 h to min



**YOU DO** On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

**ROUND 2** Questions 13–20: change decimal hours to hours and minutes, or back.**13** 6 h 24 min

---

**14** 4.2 h

---

**15** 6.75 h

---

**16** 5.5 h

---

**17** 6.4 h

---

**18** 3.2 h

---

**19** 5 h 12 min

---

**20** 6.5 h

---

**Level 3 · Challenge**

Challenging · 3 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.**Example 3** Add and subtract times.**a** 6 h 45 min + 3 h 20 min

Your teacher works this out. Copy every step.

**b** 2 h 19 min – 1 h 50 min

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Problems.

3 QUESTIONS

**REMEMBER** Add: add minutes, carry 60 min as 1 h.**1** A movie starts at 7:35 p.m. and runs for 2 h 18 min. When does it end?**2** Three songs last 3 min 42 s, 4 min 15 s and 2 min 58 s. Find the total time.**3** A runner's lap times are 1 min 52 s and 2 min 07 s. Find the difference.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill**Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Convert 150 min to hours and minutes.

**Level 2**

Evaluate 2 h 40 min + 1 h 35 min.

**Level 3**

Evaluate 5 h 10 min – 2 h 45 min.

 Extension · Stretch yourself Worked example

How many seconds are there in a week?

1.  $60 \times 60 \times 24 \times 7$

Answer: 604 800 s

**YOU DO**

On your own: read the worked example, then try these.

**E1** About how many hours old are you?

**E2** Convert 1 000 000 seconds to days, hours, minutes and seconds.

**Lesson summary**

Missed the lesson? Start here.

**Remember** 1.  $60 \text{ s} = 1 \text{ min}$ ,  $60 \text{ min} = 1 \text{ h}$ ,  $24 \text{ h} = 1 \text{ day}$ .  
3. Add: add minutes, carry 60 min as 1 h.

2. Decimal hours:  $0.1 \text{ h} = 6 \text{ min}$ .  
4. Subtract: borrow 1 h as 60 min.

 Level 1

316 min

1.  $316 \div 60 = 5 \text{ r } 16$

Answer: 5 h 16 min

 Level 2

6 h 45 + 3 h 20

1. 9 h 65 min

Answer: 10 h 5 min

 Level 3

2 h 19 – 1 h 50

1. 1 h 79 – 1 h 50

Answer: 29 min

 Extension

Seconds in a week

1.  $60 \times 60 \times 24 \times 7$

Answer: 604 800

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
12.09

# 12-hour and 24-hour time

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- convert between 12-hour and 24-hour time

## Success criteria

I can:

- ☐ use a.m. and p.m. correctly
- ☐ write 24-hour time with 4 digits
- ☐ convert midnight and noon times correctly

**Key terms** Write the definition as your teacher explains it.

a.m.

p.m.

24-hour time

Notes

## Level 1 · Getting started

Easy · 32 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Convert to 12-hour time.

**a** Convert 15:50 to 12-hour time.

**b** Convert 01:48 to 12-hour time.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

## YOU DO

On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

**REMEMBER** 24-hour time uses 4 digits: 00:00 to 23:59.

**ROUND 1** Questions 1–16: write in 24-hour time.

1 11:05 p.m.

2 7:45 p.m.

3 7:15 p.m.

4 8:05 a.m.

5 12:50 p.m.

6 2:10 p.m.

7 3:50 a.m.

8 7:25 a.m.

9 9:00 a.m.

10 4:35 a.m.

11 3:00 a.m.

12 2:25 a.m.

13 2:40 p.m.

14 12:50 a.m.

15 4:50 a.m.

16 12:20 a.m.



**YOU DO** On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

**ROUND 2** Questions 17–22: what is the time after the minutes given? Use 24-hour time.

17 125 min after 06:05

\_\_\_\_\_

18 10 min after 14:20

\_\_\_\_\_

19 150 min after 23:50

\_\_\_\_\_

20 50 min after 07:25

\_\_\_\_\_

21 85 min after 06:15

\_\_\_\_\_

22 110 min after 09:05

\_\_\_\_\_

**Level 3 · Challenge**

Challenging · 3 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Convert, then explain.

**a** Convert 12:19 a.m. to 24-hour time.

\_\_\_\_\_

Your teacher works this out. Copy every step.

**b** Why do airlines and hospitals use 24-hour time?

\_\_\_\_\_

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Problems.

3 QUESTIONS

**REMEMBER** 12:xx a.m. is 00:xx; 12:xx p.m. is 12:xx.

1 A flight leaves at 22:40 and lands 3 h 35 min later. Write the landing time in 24-hour and 12-hour time.

\_\_\_\_\_

2 A shift starts at 18:30 and ends at 02:15. How long is it?

\_\_\_\_\_

3 Write the time 20 minutes before midnight in 24-hour time.

\_\_\_\_\_

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill** Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Convert 17:25 to 12-hour time.

\_\_\_\_\_

**Level 2**

Convert 8:05 p.m. to 24-hour time.

\_\_\_\_\_

**Level 3**

Convert 12:45 a.m. to 24-hour time.

\_\_\_\_\_

 Extension · Stretch yourself Worked example

A clock shows 09:40. What time will it be 1000 minutes later?

1. 1000 min = 16 h 40 min
2. 09:40 + 16:40 = 26:20 → 02:20 the next day

Answer: 02:20 (2:20 a.m.)

**YOU DO**

On your own: read the worked example, then try these.

- E1** What time is 500 minutes after 21:15?

- E2** How many minutes are there from 07:35 to 16:10?

**Lesson summary**

Missed the lesson? Start here.

**Remember**

1. 24-hour time uses 4 digits: 00:00 to 23:59.
2. p.m. times (except 12) add 12 hours.
3. 12:xx a.m. is 00:xx; 12:xx p.m. is 12:xx.
4. Subtract 12 from 13:00 onwards for p.m.

 Level 1

15:50

1. 15 - 12 = 3

Answer: 3:50 p.m.

 Level 2

01:48

1. Before 12, a.m.

Answer: 1:48 a.m.

 Level 3

2:10 p.m.

1. 2 + 12 = 14

Answer: 14:10

 Extension

1000 min after 09:40

1. + 16 h 40 min

Answer: 02:20

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
12.10

# Time differences

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- calculate time differences and ages

## Success criteria

I can:

- ☐ count on from the earlier time to the next hour, then to the later time
- ☐ work across midnight
- ☐ calculate an age in years, months and days

**Key terms** Write the definition as your teacher explains it.

**Elapsed time**

**Time difference**

Notes

## Level 1 · Getting started

Easy · 24 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the time difference.

**a** Find the time from 5:25 a.m. to 9:45 a.m.

**b** Find the time from 11:55 p.m. to 7:30 a.m.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS**

**REMEMBER** Count on to the next hour.

**ROUND 1** Questions 1–12: find the time difference.

**1** 11:45 a.m. to 7:45 p.m.

**2** 1:20 p.m. to 6:05 p.m.

**3** 8:35 a.m. to 10:00 a.m.

**4** 7:55 a.m. to 2:00 p.m.

**5** 10:50 a.m. to 2:40 p.m.

**6** 10:05 a.m. to 5:50 p.m.

**7** 12:20 p.m. to 4:45 p.m.

**8** 8:50 a.m. to 1:25 p.m.

**9** 10:10 a.m. to 12:10 p.m.

**10** 7:05 a.m. to 12:30 p.m.

**11** 7:45 a.m. to 11:05 a.m.

**12** 10:35 a.m. to 3:55 p.m.

**YOU DO**

On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

**ROUND 2**

Questions 13–24: find the time difference (24-hour time).

13 01:25 to 03:15

\_\_\_\_\_

14 07:35 to 12:45

\_\_\_\_\_

15 04:15 to 09:25

\_\_\_\_\_

16 16:00 to 18:00

\_\_\_\_\_

17 16:25 to 16:55

\_\_\_\_\_

18 09:05 to 10:10

\_\_\_\_\_

19 14:30 to 18:35

\_\_\_\_\_

20 18:10 to 19:25

\_\_\_\_\_

21 05:40 to 06:05

\_\_\_\_\_

22 20:45 to 23:05

\_\_\_\_\_

23 02:35 to 06:40

\_\_\_\_\_

24 05:35 to 06:10

\_\_\_\_\_



**Level 2 · Building up**

Medium · 18 questions

**WE DO**

With your teacher: copy what they write on the board.

**Example 2**

Find the time difference across midnight.

**a** Find the time from 07:50 to 14:20.

Your teacher works this out. Copy every step.

**b** Find the time from 23:45 to 00:15.

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

18 QUESTIONS

**REMEMBER** Then count whole hours.

**ROUND 1**

Questions 1–12: find the time difference. It goes past midnight.

1 22:25 to 01:15

\_\_\_\_\_

2 23:00 to 00:15

\_\_\_\_\_

3 21:05 to 02:00

\_\_\_\_\_

4 22:20 to 00:55

\_\_\_\_\_

5 22:45 to 04:35

\_\_\_\_\_

6 23:25 to 02:55

\_\_\_\_\_

7 22:35 to 02:45

\_\_\_\_\_

8 22:30 to 03:35

\_\_\_\_\_

9 21:55 to 04:20

\_\_\_\_\_

10 22:25 to 00:40

\_\_\_\_\_

11 22:30 to 02:35

\_\_\_\_\_

12 23:10 to 03:35

\_\_\_\_\_

**YOU DO** On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

**ROUND 2** Questions 13–18: time zones: Sydney is 2 hours ahead of Perth. Find the time.

13 12:00 p.m. in Perth. Sydney?

\_\_\_\_\_

14 8:45 a.m. in Perth. Sydney?

\_\_\_\_\_

15 3:45 p.m. in Sydney. Perth?

\_\_\_\_\_

16 6:40 p.m. in Perth. Sydney?

\_\_\_\_\_

17 7:55 a.m. in Perth. Sydney?

\_\_\_\_\_

18 8:10 a.m. in Sydney. Perth?

\_\_\_\_\_

**Level 3 · Challenge**

Challenging · 3 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Find the age or the number of days.

**a** Samantha was born on 3 June 2004. Find her age on 16 August 2023.

\_\_\_\_\_

Your teacher works this out. Copy every step.

**b** How many days from 25 March to 12 April?

\_\_\_\_\_

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Problems.

3 QUESTIONS

**REMEMBER** Then the extra minutes. Add them up.

1 A flight leaves Sydney at 9:50 p.m. and arrives in Perth at 12:45 a.m. local time. Perth is 2 h behind Sydney. How long is the flight?

\_\_\_\_\_

2 A baby was born on 28 February 2024. How old is she (in days) on 1 April 2024?

\_\_\_\_\_

3 Jack is 12 years 7 months old. His sister is 3 years 9 months younger. How old is she?

\_\_\_\_\_

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Find the time from 8:20 a.m. to 11:05 a.m.

\_\_\_\_\_

**Level 2**

Find the time from 22:30 to 01:10.

\_\_\_\_\_

**Level 3**

How many days from 20 May to 4 June?

\_\_\_\_\_

 Extension · Stretch yourself Worked example

It is 10:00 a.m. in Sydney. What time is it in London (9 hours behind)?

1.  $10:00 - 9 \text{ h} = 1:00$

Answer: 1:00 a.m.

## YOU DO

On your own: read the worked example, then try these.

- E1 A video call is at 7:00 p.m. in Sydney. What time is it in London?

- E2 Tokyo is 1 hour behind Sydney. A plane leaves Tokyo at 20:30 local time and takes 9 h 30 min. When does it land (Sydney time)?

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Count on to the next hour.
  2. Then count whole hours.
  3. Then the extra minutes. Add them up.
  4. For ages: years, then months, then days.

 Level 1

5:25 to 9:45 a.m.

1.  $35 \text{ min} + 3 \text{ h} + 45 \text{ min}$

Answer: 4 h 20 min

 Level 2

11:55 p.m. to 7:30 a.m.

1.  $5 \text{ min} + 7 \text{ h } 30 \text{ min}$

Answer: 7 h 35 min

 Level 3

Samantha's age

1. 3 Jun 2004 → 3 Aug 2023 → 16 Aug

Answer: 19 y 2 m 13 d

 Extension

10 a.m. Sydney

1.  $- 9 \text{ h}$

Answer: 1:00 a.m.

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
12.11

## Timetables

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- read and use timetables

## Success criteria

I can:

- ☐ read departure and arrival times
- ☐ find travel times by subtracting
- ☐ choose the right service to arrive on time

**Key terms** Write the definition as your teacher explains it.**Timetable****Departure**

Notes

## Level 1 · Getting started

Easy · 20 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Use the train timetable.

| Station          | a.m. | a.m. | a.m.  | p.m.  |
|------------------|------|------|-------|-------|
| Moss Vale        | 6:05 | 7:40 | 9:10  | 7:45  |
| Bowral           | 6:12 | 7:47 | 9:17  | 7:52  |
| Mittagong        | 6:20 | 7:55 | 9:25  | 8:00  |
| Picton           | 6:58 | 8:33 | 10:03 | 8:38  |
| Campbelltown     | 7:25 | 9:00 | 10:30 | 9:05  |
| Sydney (Central) | 8:21 | 9:56 | 11:26 | 10:01 |

**a** When does the 7:40 a.m. train reach Picton?

Your teacher works this out. Copy every step.

**b** How long does the 7:45 p.m. train take to get to Sydney?

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **20 QUESTIONS****REMEMBER** Find the row for the station and the column for the service.**ROUND 1** Questions 1–12: how long is each trip?**1** 11:15 → 13:20**2** 09:30 → 12:50**3** 08:55 → 10:05**4** 11:40 → 11:55**5** 08:25 → 09:25**6** 13:25 → 15:10**7** 16:30 → 17:05**8** 15:50 → 16:40**9** 10:05 → 12:35**10** 08:10 → 10:50**11** 15:15 → 16:15**12** 14:45 → 16:00

## YOU DO

On your own: **Easy** keep going: Round 2.

8 QUESTIONS MORE

## ROUND 2

Questions 13–20: use the bus timetable.

| Stop    | Bus 1 | Bus 2 | Bus 3 | Bus 4 |
|---------|-------|-------|-------|-------|
| School  | 08:00 | 08:35 | 09:00 | 09:50 |
| Library | 08:07 | 08:42 | 09:07 | 09:57 |
| Pool    | 08:19 | 08:54 | 09:19 | 10:09 |
| Shops   | 08:28 | 09:03 | 09:28 | 10:18 |
| Beach   | 08:42 | 09:17 | 09:42 | 10:32 |

- 13 When does Bus 1 reach the Pool?

---

- 14 When does Bus 2 reach the Shops?

---

- 15 When does Bus 3 reach the Beach?

---

- 16 When does Bus 4 reach the Library?

---

- 17 How long from School to Beach?

---

- 18 How long from Library to Shops?

---

- 19 How long from Pool to Beach?

---

- 20 How long from School to Pool?

---

Level 2 · Building up

Medium · 15 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Use the train timetable.

| Station          | a.m. | a.m. | a.m.  | p.m.  |
|------------------|------|------|-------|-------|
| Moss Vale        | 6:05 | 7:40 | 9:10  | 7:45  |
| Bowral           | 6:12 | 7:47 | 9:17  | 7:52  |
| Mittagong        | 6:20 | 7:55 | 9:25  | 8:00  |
| Picton           | 6:58 | 8:33 | 10:03 | 8:38  |
| Campbelltown     | 7:25 | 9:00 | 10:30 | 9:05  |
| Sydney (Central) | 8:21 | 9:56 | 11:26 | 10:01 |

**a** What is the latest train you can catch at Moss Vale to be in Sydney by 11 a.m.?

Your teacher works this out. Copy every step.

**b** How long is the trip from Bowral to Campbelltown on the 9:10 a.m. train?

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

15 QUESTIONS

**REMEMBER** Read across or down carefully.

**ROUND 1** Questions 1–15: how long is each trip? It goes past the hour.

**1** 8:50 a.m. → 11:05 a.m.

\_\_\_\_\_

**2** 10:45 a.m. → 1:10 p.m.

\_\_\_\_\_

**3** 2:40 p.m. → 4:35 p.m.

\_\_\_\_\_

**4** 10:50 a.m. → 1:20 p.m.

\_\_\_\_\_

**5** 11:55 a.m. → 12:15 p.m.

\_\_\_\_\_

**6** 7:55 a.m. → 8:50 a.m.

\_\_\_\_\_

**7** 5:50 p.m. → 7:25 p.m.

\_\_\_\_\_

**8** 10:45 a.m. → 11:45 a.m.

\_\_\_\_\_

**9** 1:45 p.m. → 3:50 p.m.

\_\_\_\_\_

**10** 8:55 a.m. → 10:25 a.m.

\_\_\_\_\_

**11** 12:40 p.m. → 3:05 p.m.

\_\_\_\_\_

**12** 11:45 a.m. → 12:45 p.m.

\_\_\_\_\_

**13** 12:50 p.m. → 1:40 p.m.

\_\_\_\_\_

**14** 11:55 a.m. → 12:25 p.m.

\_\_\_\_\_

**15** 3:55 p.m. → 6:00 p.m.

\_\_\_\_\_



Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Use the train timetable, then explain.

| Station          | a.m. | a.m. | a.m.  | p.m.  |
|------------------|------|------|-------|-------|
| Moss Vale        | 6:05 | 7:40 | 9:10  | 7:45  |
| Bowral           | 6:12 | 7:47 | 9:17  | 7:52  |
| Mittagong        | 6:20 | 7:55 | 9:25  | 8:00  |
| Picton           | 6:58 | 8:33 | 10:03 | 8:38  |
| Campbelltown     | 7:25 | 9:00 | 10:30 | 9:05  |
| Sydney (Central) | 8:21 | 9:56 | 11:26 | 10:01 |

a Which train arrives in Sydney closest to 10 a.m.?

Your teacher works this out. Copy every step.

b Why are timetables often written in 24-hour time?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Make your own timetable.

2 QUESTIONS

REMEMBER Travel time = arrival – departure.

- 1 A ferry leaves the wharf every 40 minutes from 06:50 until 09:30. The trip takes 25 minutes. Write a timetable showing departure and arrival times.

| Departs | Arrives |
|---------|---------|
|         |         |
|         |         |
|         |         |
|         |         |
|         |         |
|         |         |

- 2 Your school day as a timetable: list the start and end of each period in 24-hour time.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

You live 10 minutes' walk from Bowral station. You must be in Sydney by 10:15 a.m. What time must you leave home?

1. The 7:40 a.m. train arrives at 9:56 a.m.
2. It leaves Bowral at 7:47 a.m.
3. Leave home 10 min earlier

Answer: 7:37 a.m. (or earlier)

YOU DO

On your own: read the worked example, then try these.

- E1 Your friend at Picton wants to meet you on the same train. Which time does she board?

- E2 The 7:40 a.m. train is running 12 minutes late. When will it reach Sydney?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skill

Choose ONE question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

Level 1

When does the 9:10 a.m. train reach Campbelltown?

Level 2

How long is the bus trip from School to Shops?

Level 3

Which train should you catch from Moss Vale to reach Sydney before 9 a.m.?

## Lesson summary

Missed the lesson? Start here.

### Remember

1. Find the row for the station and the column for the service.
2. Read across or down carefully.
3. Travel time = arrival – departure.
4. Choose the service that arrives before you need to be there.

#### Level 1

7:45 p.m. to Sydney

1. 7:45 p.m. → 10:01 p.m.

Answer: 2 h 16 min

#### Level 2

In Sydney by 11 a.m.

1. 9:10 arrives 11:26 (too late)

Answer: 7:40 a.m.

#### Level 3

Library to Beach

1. 15:18 → 15:45

Answer: 27 min

#### Extension

Leave home for Bowral

1. 7:47 – 10 min

Answer: 7:37 a.m.

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.